

Quick Recap of Basics

Let's quickly recall what these measures do.

- **Mean** → The arithmetic average; it represents the *center* of the data.
- **Median** → The middle value when data are sorted.
- **Standard Deviation (SD)** → Measures how far data points typically spread from the mean.
- **Interquartile Range (IQR)** → The difference between the 75th and 25th percentiles — in other words, it captures the *middle 50%* of your data.

Both pairs measure **center** and **spread**, but they behave very differently when your data has **outliers** or **skewness**.

Why Mean and SD Fail Sometimes

Mean and standard deviation work great for **symmetric, normally distributed data** — like heights or IQ scores in large populations.

But if you have **outliers** — for example, one billionaire in an income dataset — the mean gets pulled toward the extreme.

Example:

```
import numpy as np
data = [25, 27, 28, 29, 30, 31, 35, 500] # one big outlier
print("Mean:", np.mean(data))
print("Median:", np.median(data))
```

The mean shoots up, but the median stays stable — that's why we call median a **robust** measure.

When to Use Median and IQR

Use **Median and IQR** when:

1. **Data are skewed or have long tails**
 - Example: *Income, house prices, YouTube views, medical costs*.
 - These are **right-skewed**, and the mean gets dragged upward by a few large values.
 - The median gives a better picture of a "typical" case.
2. **There are outliers or extreme values**
 - Outliers inflate SD and make the spread look huge even if most data are normal.
 - IQR ignores extremes by focusing on the middle 50%.
3. **You need robust statistics for ML preprocessing**
 - Algorithms like **RobustScaler** in sklearn use **median and IQR** instead of mean and SD to scale data safely.
 - It's perfect for datasets where outliers shouldn't dominate scaling.

When Mean and SD Are Still Better

Use **Mean and SD** when:

- Data are roughly **normal (bell-shaped)**.

- You plan to apply algorithms assuming normality — for instance, **Linear Regression**, **Naive Bayes**, or **Z-score scaling**.

Wrap-up and Summary

So to summarize:

- **Mean & SD** → Best for *normal, clean data*.
- **Median & IQR** → Best for *skewed, messy, or outlier-heavy data*.

Always visualize first — a simple histogram or boxplot can tell you whether your data needs a “robust” approach or not.